# Yeruva Venkata Vamsi Reddy

ep24btech11034@iith.ac.in | yvenkatavamsireddy@gmail.com | +91 7013868309 | LinkedIn | GitHub

## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 7.13 |
| Narayana Junior College | Class XII (Telangana State Board of Intermediate Education) | 2024 | 9.41% |
| Bhashyam High School | Class X (Board of Secondary Education, Telangana) | 2022 | 10.00% |

## PROJECTS

**Scribble-Game | PyTorch, FastAPI, Python, HTML, CSS, JavaScript**

- Developed an AI-powered sketch recognition web application using a **multi-layer Convolutional Neural Network (CNN)** trained on **15,000+ images per class** from the Google Quick, Draw! dataset.
- Built an end-to-end inference pipeline with image preprocessing, GPU-accelerated training, and Top-3 real-time predictions across N object categories using PyTorch.
- Designed a FastAPI backend and HTML5 Canvas frontend, enabling users to draw sketches on a **280×280** canvas and receive predictions from a model trained on **100,000+** sketches.

**CodeVerdict | Full-Stack Online Judge Platform**

Tech Stack: Python, FastAPI, HTML, CSS, JavaScript, SQLite, SQLAlchemy, Docker, JWT

- Developed a full-stack online coding platform for secure code submission and automated evaluation, inspired by LeetCode and Codeforces.
- Built RESTful APIs with **FastAPI**, implemented **JWT authentication**, and developed a responsive frontend using HTML, CSS, and JavaScript.
- Engineered a **Docker-based sandbox** to securely execute user code with resource limits and generate AC/WA/TLE/RE verdicts.

**URL Shortener | Backend Web Application**

Tech Stack: Python, FastAPI, SQLite, Pydantic

- Developed a URL shortening service using FastAPI and SQLite with automatic redirection and persistent URL storage.
- Implemented 30-day URL expiration, reuse of existing active links, and collision detection for reliable short-code generation.
- Designed RESTful APIs with interactive **Swagger UI** documentation for URL creation and redirection.

## POSITIONS OF RESPONSIBILITY

**Head of Logistics, Media Council**

- Promoted from Logistics Coordinator to Head of Logistics in recognition of consistent contributions, ownership, and leadership within the Media Council.
- Leading the planning and execution of logistics for council events by coordinating cross-functional teams, allocating resources, and ensuring seamless on-ground operations.
- Managing volunteers and collaborating with multiple stakeholders to optimize event workflows, delegate responsibilities effectively, and deliver successful events under tight deadlines.

**Coordinator, Infrastructure & Security Domain, Elan & nVision Club**

- Assisted in infrastructure setup and security operations for large-scale technical and cultural events, ensuring venue readiness and smooth event execution.

- Coordinated with organizing teams and volunteers to manage equipment deployment, access control, crowd management, and on-ground operational support.

## SKILLS

Programming Languages: Python, C++, C, SQL

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems.

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Prompt Engineering

Backend Development: FastAPI, SQLite, SQLAlchemy, Authentication (JWT)

Libraries & Frameworks: NumPy, Pandas, Matplotlib, PyTorch, Scikit-learn

Databases: SQLite, MySQL

Developer Tools: Git, GitHub, PyCharm, VS Code, Docker, Siemens Solid Edge

## RELEVANT COURSES

Computer Science : Data Structures & Application, C Programming, Introduction to Artificial Intelligence, Computational Physics

Mathematics : Probability & Statistics, Linear Algebra, Differential Equations, Calculus

## EXTRACURRICULAR

Sports: Badminton, Cricket, Volleyball

A part of National Service Scheme(NSS)